



INTERNSHIP & JOB PLACEMENT HUB

OFFICIAL DOCUMENTATION

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Project Specifics

This project is based on addressing the problem of connecting employers with prospective employees. The aim of the project was to create a system that employers can enlist their job posts and for job seeking individuals to be able to see the jobs enlisted and apply for the job(s) that match their skill set or degree.

1. User Manual

This user manual provides guidance on how to use the Job and Internship Portal effectively. It is designed to help students, job seekers, and employers navigate the system with ease, from account registration to applying for opportunities or posting available positions. The manual explains each feature step-by-step to ensure users can make the most of the platform and achieve their goals efficiently.

1.1 System Overview

The Job and Internship Portal is an online platform that connects NWU's final-year students and graduates with potential employers and internship providers. It allows users to create profiles, upload resumes, search for available opportunities, and apply directly through the system. Employers can also register, post job and internship listings, and manage applications received. The portal provides an easy and efficient way for both parties to interact and meet their recruitment or career goals.

1.1.1 System Navigation / Dashboard Overview

Once users have logged in, they are directed to their dashboard. The dashboard provides a central location where users can access all features relevant to their role, whether they are an applicant or an employer.

Applicant Dashboard

- The dashboard displays available jobs and internships, with search and filter options at the top.
- Users can access their personal profile to update information or upload resumes.
- A section is dedicated to tracking the status of applications submitted.

Employer Dashboard

- Employers can see an overview of all active job and internship postings.
- Quick access to creating new job postings and managing existing listings is available.
- Applications submitted by job seekers are listed for review, and employers can shortlist candidates.

Menu and Navigation

- The main menu or sidebar provides easy access to all key sections such as Jobs, Applications, and Profile.
- Users can click on any menu item to quickly move between sections without losing their current work.
- Logout is available at the top-right corner to securely exit the system.

This overview helps users understand the layout and easily find the features they need to perform their tasks efficiently.

1.2 Getting started

To begin using the Job and Internship Portal, you will need an internet connection and a web browser. Open the portal by entering the provided web address in your browser's address bar. Once the homepage loads, you can choose to either register a new account or log in if you already have one.

Follow these steps to get started:

1.2.1 Go to the portal's homepage.

When the homepage opens, you will see an About Us section. As you scroll down, there is a short paragraph that introduces the system and explains its purpose. This helps you understand what the portal is designed for before you register or log in.

[Home](#)[Find Jobs](#)[Contact](#)[My Profile](#)[My Applications](#)[Employer Admin](#)[Register](#)[Login](#)

About Us

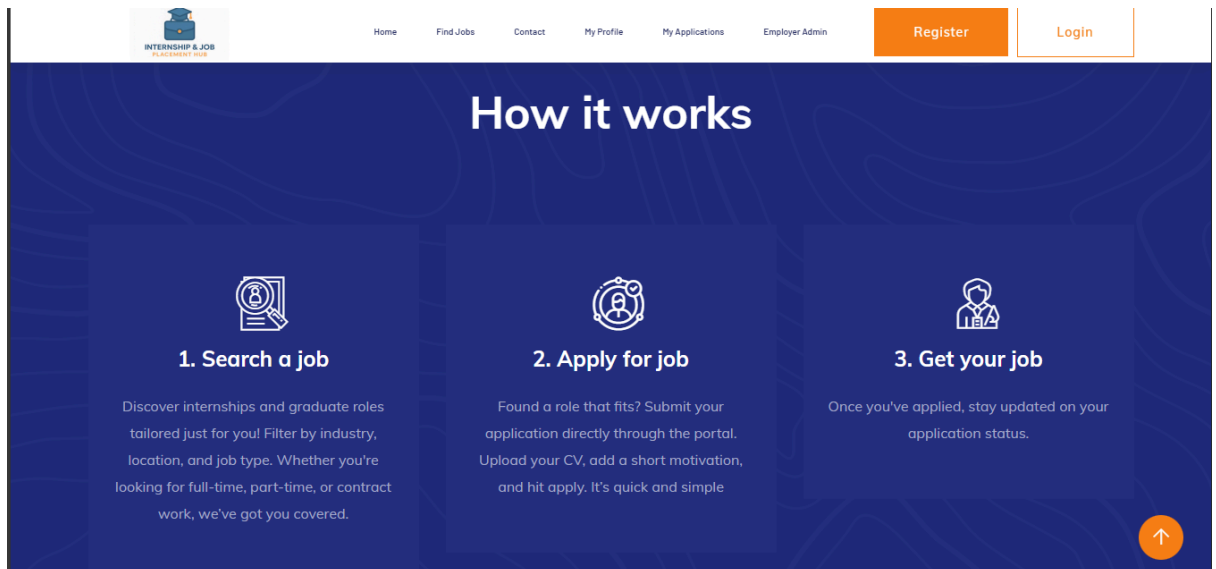
[Home](#)[Find Jobs](#)[Contact](#)[My Profile](#)[My Applications](#)[Employer Admin](#)[Register](#)[Login](#)

WHAT WE ARE DOING

Internship & Job Portal

We are building a platform that connects students, graduates, and professionals with meaningful career opportunities. Our goal is to make internships and jobs more accessible by bringing applicants and employers together in one simple, transparent, and user-friendly space. Whether you are looking to gain hands-on experience, kick-start your career, or find the right talent for your organization, our portal makes the process faster, smarter, and easier for everyone.





1.2.2 Click on the “Register” button to create a new account.

- Fill in your personal details such as name, username, email address, and password.
- Select your user type (Job Seeker or Employer).
- Submit the registration form.
- Once registered, log in using your email and password to access your dashboard.

1.2.3.If you are already registered on the system

Click on the “Login” button to sign in using the username and password you created during registration.

Please enter your details

Welcome Back

Username

Password

☐ Remember me [Forgot password?](#)

Sign In

Or



Don't have an account? [Sign up](#)

1.3 User Roles and Features

The Job and Internship Portal has two main types of users: Applicants which are the Job Seekers and Employers. Each user type has access to specific features designed to meet their needs. After logging in, users are directed to a personalized dashboard where they can perform tasks related to their role.

1.3.1 Applicant's Perspective

From the applicant's perspective, the system allows job seekers or students to perform various actions after logging in.

- 1. After the applicant has logged in, they can find a job by searching using a keyword or job title, or by filtering results using the location dropdown menu.**

Find the most exciting startup jobs



2. Browse top categories in which the jobs fall under.

Browse Top Categories



Design & Creative
(653)



Design & Development
(658)



Sales & Marketing
(658)



Mobile Application
(658)



Construction
(658)



Information Technology
(658)



Real Estate
(658)



Content Writer
(658)



Design & Creative
(653)



Design & Development
(658)



Sales & Marketing
(658)



Mobile Application
(658)



Construction
(658)



Information Technology
(658)

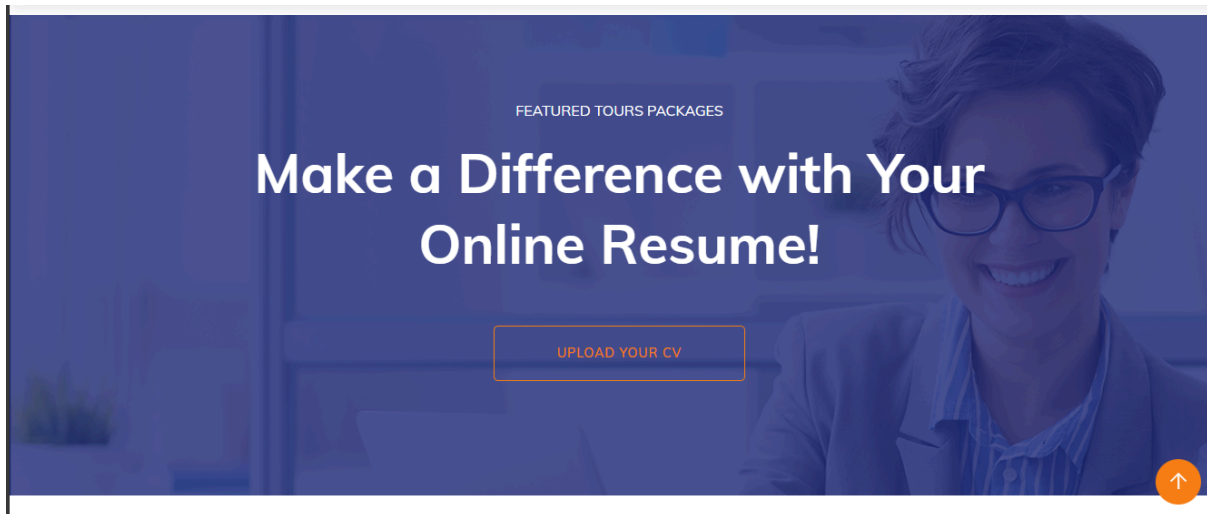


Real Estate
(658)

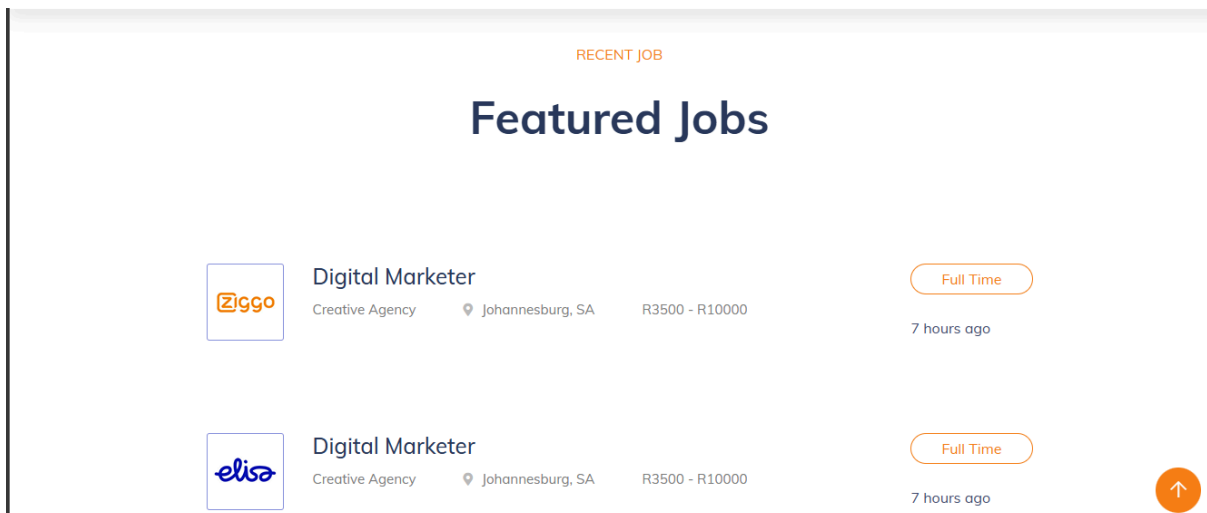


Content Writer
(658)





3. As they scroll down, applicants can view the latest jobs that have been posted.



4. Once they click a job they are interested in, they can apply for the position directly through the portal.
5. On their personal profile, applicants can upload or update their resume and profile information.

Settings

Profile Settings

Account

Password

Public info

Username

Username

Biography

Tell something about yourself

Save Changes



Upload

For best results, use an image at least 128px by 128px in .jpg format

Private info

First name

First name

Last name

Last name

Email

Email

Email

Address

1234 Main St

Address 2

Apartment, studio, or floor

City

Province

Zip

Save Changes

↑

6. Applicants can track the status of submitted applications.

Student Profile

Dashboard

Logout

2 Applications

Recent Applications

Filter by Status

Filter by Job Applied

Apply Filter

Action

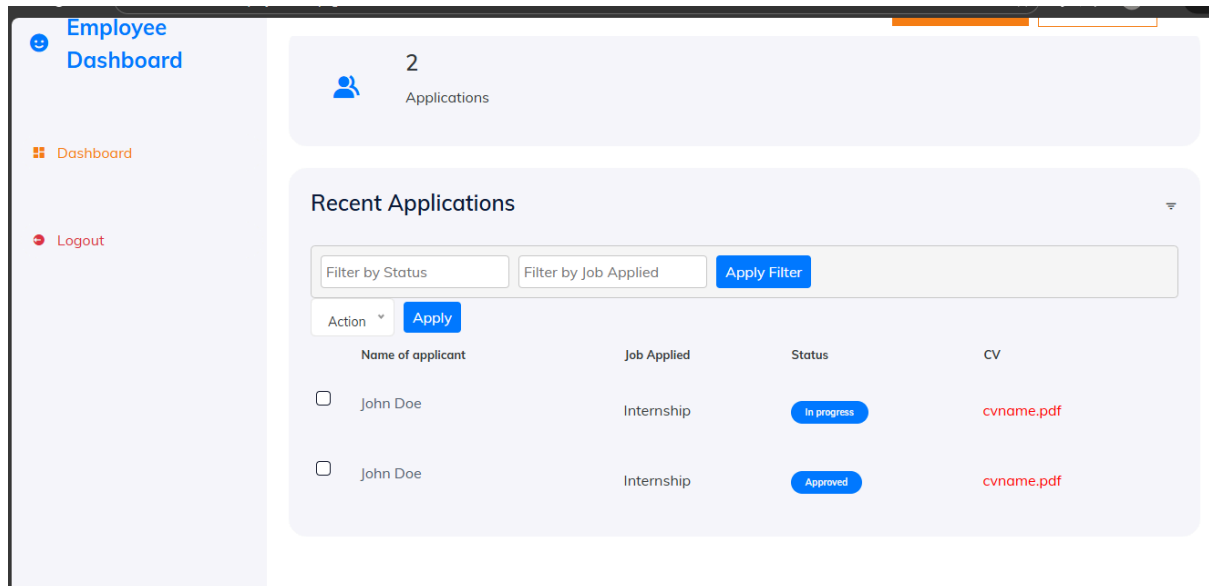
Apply

Job applied	Company	Status	CV	Withdraw Application
☐ Internship	Tech Creators	In progress	cvname.pdf	Withdraw Application
☐ Internship	Amazing Tech	In progress	cvname.pdf	Withdraw Application

1.3.2 Employee perspective

From the employer's perspective, the system allows organizations or recruiters to perform various actions after logging in.

1. **After logging in, employers can access their dashboard to view all current job and internship postings.**



2. **Post new job or internship listings with details such as job title, description, location, and application**

Job Details

Job Title

Job Type

Job Description

Job Location

Number of Vacancies

Salary Range

Required Knowledge, Skills and Abilities

Skills Required

Education + Experience

Experience/Education

☐ Background check consent

2. Front End Team Reflections

The front-end development of the Job and Internship Portal was a significant part of the project that tested our technical abilities as well as our teamwork, problem-solving, and time management skills. Throughout the process, the team faced various challenges, made important decisions, and worked collaboratively to ensure the portal was functional, user-friendly, and visually appealing. This section reflects on the difficulties encountered and identifies prospective improvements for future projects.

2.1 Challenges encountered

During the front-end development, the team encountered several technical and organizational challenges that required careful planning, collaboration, and problem-solving.

1. Limited experience with React JS

Our team initially planned to develop the front end using React JS. However, due to our limited experience with the framework, implementing features efficiently became difficult. After discussing the situation, we collectively decided to switch to HTML, which allowed us to work more effectively and meet deadlines without compromising the quality of the portal.

2. Time management and scheduling conflicts

Coordinating development alongside academic responsibilities, including tests and other coursework, sometimes disrupted our schedule. The team had to carefully plan and manage time to ensure consistent progress while balancing these external pressures.

3. Designing a user-friendly and appealing layout

Choosing the right layout, color schemes, and overall design to make the portal both visually appealing and intuitive for users was challenging. It required multiple iterations, discussions, and testing to finalize a design that met our standards.

4. Responsive design and browser compatibility

Ensuring that the portal worked seamlessly across various devices and screen sizes, as well as maintaining consistency across different browsers, required thorough testing and adjustments.

5. Learning new techniques while developing

While building the portal, the team often had to research and learn new HTML and CSS techniques on the go. This added pressure but also helped us grow our skills as we tackled real development challenges together.

Throughout these challenges, the team demonstrated exceptional dedication and teamwork. Members collaborated closely, dividing tasks based on individual strengths and continuously supporting each other to

solve coding and design problems. The commitment and effort from each team member were vital in overcoming obstacles and ensuring the portal was functional, user-friendly, and delivered on time.

2.2 Prospective Improvements possible

While the team managed to successfully develop the front-end of the portal, there are several areas where we identified opportunities for improvement to enhance efficiency, functionality, and overall user experience:

1. Enhance visual design and user experience.

Investing additional time in refining layouts, color schemes, and interface elements could make the portal more visually appealing and intuitive. Features such as interactive elements, tooltips, or improved form validation could further enhance usability.

2. Focus on responsive and adaptive design.

Although the portal was tested across devices, further optimization for different screen sizes and browsers could provide an even better user experience for all users.

3. Strengthen code organization and maintainability.

Structuring HTML, CSS, and JavaScript more modularly and documenting the code clearly would make future updates easier and support team members who may join later.

4. Continuous learning and skill development.

The team benefited from researching new techniques during development. Allocating time for dedicated training or practice on frameworks and best practices in advance could improve efficiency and confidence in future projects.

3. Back End - Database Team Reflections

3.1 Challenges encountered

We individually had to create tables that would store different information and we chose to use Oracle in the beginning of the project but later on used MySQL. The problem arose when we had to allow for filtering with the indexes but with clear communication and the assistance of AI we were able to solve the problem.

As database engineers, we chose to set up a server to host the system database. We used Microsoft Azure. We set up the server on Azure and connected it to VS Code to construct a live database. In VS Code, we used the PostgreSQL extension as indicated in our documentation. We constructed a separate database to store all of our data rather than using the default database offered by VS Code. This prevented conflicts while constructing tables from scratch. We merely made an empty database. The backend team was in charge of constructing the tables. Our role was to construct a live database that the backend team could easily access and use.

Advantages

1. The backend team could access the database easily.
2. The database is live, so the system is less likely to crash.
3. The database supports the tools used by the backend team.
4. Hosting on Azure ensures high availability and reliability.
5. Using a separate database reduces conflicts and errors during development.

Disadvantages

1. We had to set the IPs to zeros for the backend team to access the database, which reduces security despite strong passwords.

2. Integrating the database with the backend team took a lot of time.
3. Initial setup and linking to VS Code required careful configuration.
4. Any server downtime on Azure would affect all teams accessing the database.

More challenges that we faced include:

1. Constraints not functioning properly. Check constraints, such as email verification, were applied during the table creation process. This resulted in issues because other versions of MySQL ignored the check limitations. The best method to do this was to validate data directly on the front end.
2. Confusion over table relationships and foreign keys. While creating the applications table, the student_id accessed the users table rather than the students table, causing confusion between the two tables.
3. Difficulty exporting a complete schema (table dump) for external use.
4. Referring to previously generated tables in the database.

3.2 Prospective Improvements possible

None at this stage.

4. Back End - Coding Team Reflections

1. Tech Stack

Category	Technology / Tool	Description
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Programming Language	Python	Primary language used for backend development and logic implementation.
Framework	Django & Django REST Framework (DRF)	Used for building the web application and exposing RESTful APIs.
Database	PostgreSQL	Relational database system for structured data storage and management.
Containerization	Docker	Enables consistent development and deployment through containerized environments.
Cloud Hosting	Microsoft Azure	Platform for deploying and hosting the web application and related services.
API Testing Tool	Postman	Utilized for designing, testing, and validating API endpoints.

2. Challenges Faced

- Setting up the initial Docker environment with PostgreSQL and managing Django database migrations.
- Configuring environment variables for seamless deployment to Azure.
- Handling file uploads for CVs and resumes, including proper media storage configuration.
- Ensuring role-based access and field visibility between students and employers.
- Implementing a secure password reset mechanism using tokens and email verification.

3. Data Models

Model	Description	Key Fields / Relationships
User	Base user model extending Django's AbstractUser. Used for authentication and role management.	• Roles: Student or Employer • Common fields: username, email, password
Student	Represents a student profile linked to a user account. Stores student-specific information.	• One-to-one relationship with User • Fields: degree, year, CV (resume)
Employer	Represents an employer profile linked to a user account. Contains organization-related data.	• One-to-one relationship with User • Fields: company name, industry
Job	Job postings created by employers.	• Foreign key to Employer • Fields: title, description, job type, required experience, education, salary, location, etc.

Application	Represents a student's application to a job.	• Foreign key to Job and Student • Fields: cover letter, resume, status (pending, reviewed, shortlisted, accepted, rejected)
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User → Student: One-to-one

User → Employer: One-to-one

Employer → Job: One-to-many

Job → Application: One-to-many

Student → Application: One-to-many

4. API Endpoints

Authentication & User Management

This section covers all endpoints related to user registration, authentication, and profile management. It handles account creation, login/logout, password resets, and CV uploads for students. These APIs ensure secure user access and proper identity management.

Base URL: <http://localhost:8000/>

Method	Endpoint	Description	Access / Notes
POST	/users/register/	Register a new user (student or employer)	Public

POST	/users/login/	Authenticate user and return token	Public
POST	/users/logout/	Logout and invalidate authentication token	Authenticated users
POST	/users/reset-password/	Change password (requires old password)	Authenticated users
POST	/users/forgot-password/	Request a password reset link via email	Public
POST	/users/reset-password-confirm/	Reset password using token and UID	Public
POST	/users/upload-cv/	Upload CV or resume (students only)	Authenticated (Student)

User Management (CRUD Endpoints)

These endpoints allow the retrieval of user data for administrative or debugging purposes. They enable listing of all registered users, as well as filtering specifically for students or employers.

Method	Endpoint	Description	Access / Notes
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GET	/users/users/	Retrieve all users (for admin/debugging)	Admin only
GET	/users/students/	List all student users	Admin / Employer (optional)
GET	/users/employers/	List all employer users	Admin / Student (optional)

Job Endpoints

This section defines endpoints used to manage job postings. Employers can create, update, and delete jobs, while students and visitors can browse, search, and filter available job listings. Employers can also view statistics related to their job postings.

Base URL: /job/

Method	Endpoint	Description	Access / Notes
GET	/jobs/	List all active jobs (supports search and filtering)	Public
POST	/jobs/create/	Create a new job posting	Authenticated (Employer)
GET	/jobs/<id>/	Retrieve details of a specific job	Public
PUT	/jobs/<id>/	Update a job posting	Authenticated (Employer)

DELETE	/jobs/<id>/	Delete a job posting	Authenticated (Employer)
GET	/jobs/my/	List jobs created by the logged-in employer	Authenticated (Employer)
GET	/jobs/stats/	Retrieve job posting statistics for the employer	Authenticated (Employer)

Application Endpoints

These endpoints manage the job application process. Students can apply for jobs and track their applications, while employers can review and update the status of applications submitted to their posted jobs.

Base URL: /job/

Method	Endpoint	Description	Access / Notes
POST	/applications/create/	Apply for a job	Authenticated (Student)
GET	/applications/my/	View applications submitted by the logged-in student	Authenticated (Student)
GET	/applications/job/<job_id>/	Employer views applications	Authenticated (Employer)

		for a specific job	
PATCH	/applications/<application_id>/update-status/	Employer updates application status or notes	Authenticated (Employer)
GET	/applications/<application_id>/	View application details	Authenticated (Student or Employer)

5. Permissions

The following user permissions have been set up to control the workflow of the application based on a specific role:

Students can:

- Register/login/logout
- Upload CVs
- Apply to jobs
- View their own applications

Employers can:

- Register/login/logout
- Post jobs, edit/delete their own jobs
- View applicants for their jobs
- Update application statuses
- Access job stats

6. Docker & Deployment Notes

Dockerfile: Builds and containerizes the Django application using a Python base image.

docker-compose.yml: Orchestrates multi-container setup, connecting the Django web service to the PostgreSQL database.

Exposed Port: 8000 — the Django application is accessible at <http://localhost:8000>.

Run Command:

```
docker-compose up --build
```

This command builds the images (if not already built) and starts all services defined in the docker-compose.yml file

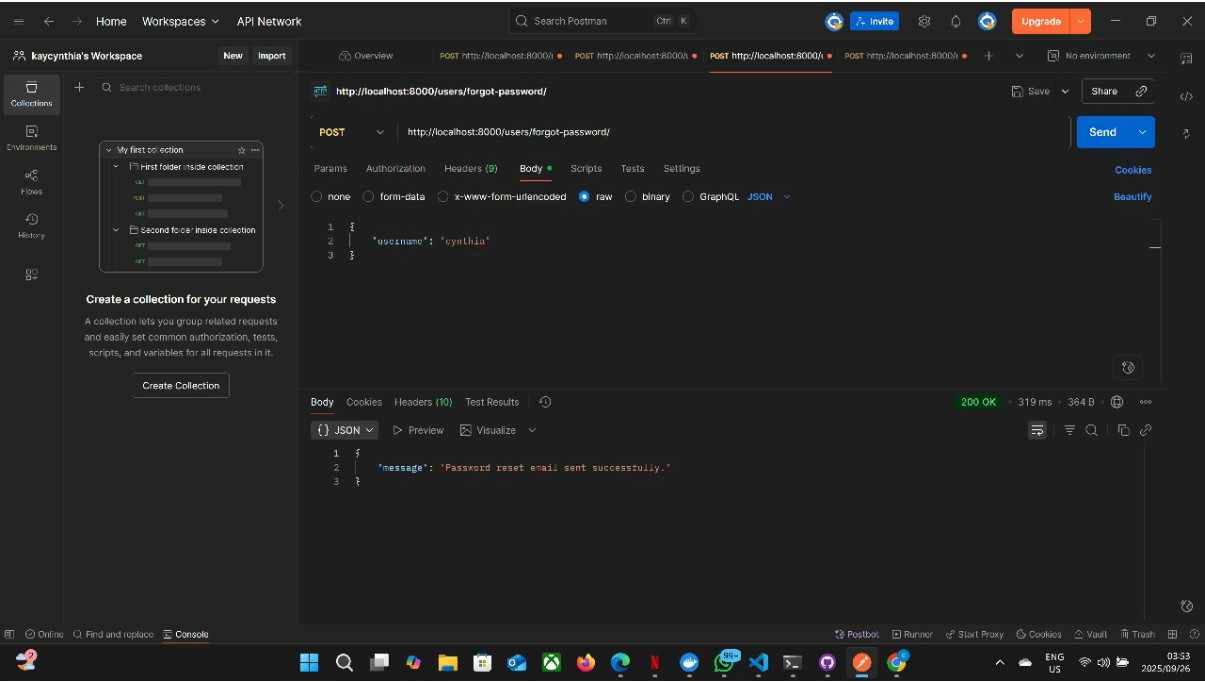
7. Postman Usage

Tool Used: Postman collections were used to test and validate all API endpoints.

Base URL: <http://localhost:8000/>

Authentication: Token-based authentication is used for all secured endpoints.

File Uploads: Endpoints that handle file uploads (e.g., CV submissions) use form-data format in Postman.



8. Advantages of Our Technology & Approach

Technology Advantages

Technology	Advantage
Django REST Framework	Provides a clean and maintainable API structure with easy role-based field control and powerful serializers for managing user data.
PostgreSQL	Offers strong relational data handling, ensuring data integrity and efficient querying for complex relationships.

Docker	Enables a consistent and reproducible environment across development, testing, and deployment stages.
Azure Hosting	Delivers a reliable and scalable cloud infrastructure with integrated CI/CD pipelines for automated deployment.
Email Integration	Implements a secure and practical password reset flow aligned with real-world application standards.
Postman Testing	Used to validate and verify key API workflows such as authentication, CV uploads, and password reset functionality.

Approach Advantages

Approach	Benefit
Role-Based Field Separation	Improves user experience and data integrity by clearly distinguishing Student and Employer data views.
Secure Password Reset Flow	Increases user trust and supports compliance with modern security standards.
Student-Centric Simplicity	Simplifies onboarding and reduces barriers to entry for new users.

Modular Design	Ensures scalability and flexibility, making it easier to introduce new roles or extend functionality in the future.
Containerized Deployment	Enhances portability, environment consistency, and ease of scaling across different platforms.

5. UAT presentation feedback

The User Acceptance Testing (UAT) session was held on **8 October 2025**. As the work was completed in four groups, the feedback was provided per team:

Front End:

The feedback was positive regarding the overall look, feel, and flow of the website. However, it was noted that the website is not yet fully integrated with the back end, which limits certain functionalities. The team committed to resolving these integration issues soon.

Back End and Database:

These teams provided a brief overview of the system design and available functionalities. No major changes were requested at this stage. It was reiterated that successful integration with the front end remains a key next step, requiring close collaboration with the full-stack and front-end developers.

Integration (Full-Stack Developers):

The team agreed that integration will be the primary focus for the remaining project period. It is essential to ensure that all front-end features function seamlessly with the back-end and database components and that the system as a whole meets user expectations.

6. Conclusion

As a team, we have been privileged to work on this project together. Most of us, learnt new technologies and frameworks along the way which can really take us further and make our grass greener going forth.

As with any team working on a project such as this, there were challenges we encountered such as distance between partners, a team member(s) not optimally contributing to the projects. Over and above all, we are happy with what we could produce, amidst the circumstances.

Our desire is for this project to make it easier for employers and job seekers to connect. And if this part is fulfilled, our job would have been a success!